

Scientific Machine Learning: Blending Science with Data

Homework Day 7

Write a Julia program to find the gradient (i.e. BOTH $\partial f/\partial x_1$ and $\partial f/\partial x_2$) for the function below

$$f(x_1, x_2) = \exp(-x_1^2) \cos(3x_2)$$

using

- a) analytical symbolic techniques (just do the differentiation by hand and write the derivative as a function in Julia)
- b) the `Zygote` package in Julia
- c) Dual numbers introduced in lectures. For this part of the question, you are meant to only define the minimal amount of operators that you need to calculate the derivative for the function above.

Compare your output for all three methods above at $(x_1, x_2) = (2.0, 3.0)$